

1-D Kinematics

SS ILP Physics (Intermediate), Week 1 Class 2



Outline

- Kinematics
- Position
- Displacement
- Velocity
- Acceleration
- Kinematics with constant acceleration

Kinematics

- Kinematics is the description of the motion of moving objects without considering the forces that cause or change the motion.
- Dynamics is the study of the motion of moving objects and forces.

<https://en.wikipedia.org/wiki/Kinematics>
[https://en.wikipedia.org/wiki/Dynamics_\(mechanics\)](https://en.wikipedia.org/wiki/Dynamics_(mechanics))

Position

- Position is defined as the location of an object.
- The position of an object is described by a vector in a coordinate system, usually varying with time $\vec{r}(t)$.
- SI unit: meter (m).

<https://courses.lumenlearning.com/atd-monroecc-physics/chapter/chapter-1/>

Distance & Displacement

- Distance is defined as the length of the path travelled by a moving object. SI unit: meter (m).
- Displacement is defined as the change in position. SI unit: meter (m).

[https://phys.libretexts.org/Bookshelves/College_Physics/Book%3A_College_Physics_\(OpenStax\)/02%3A_Kinematics/2.01%3A_Displacement](https://phys.libretexts.org/Bookshelves/College_Physics/Book%3A_College_Physics_(OpenStax)/02%3A_Kinematics/2.01%3A_Displacement)

Example (1)

You need to walk 1 km to the east, followed by 2 km to the north to reach your school from home. Find

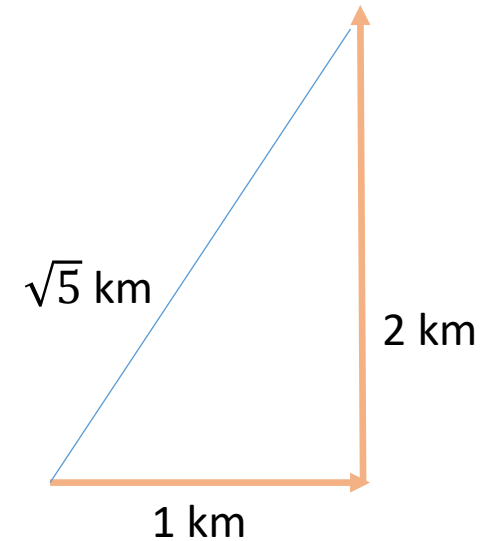
- a) the distance travelled, and
- b) the displacement.

Ans:

a) Distance = 3 km

b) Displacement: $\Delta\vec{r} = 1(\text{km})\hat{i} + 2(\text{km})\hat{j}$

Magnitude of displacement is $\Delta r = \sqrt{5} \text{ km}$



Average velocity

- Average velocity is defined as:

$$\text{average velocity} = \frac{\text{total displacement}}{\text{total time taken}} = \frac{\Delta \vec{r}}{\Delta t}$$

- SI unit: meter/second (m/s).

[https://phys.libretexts.org/Bookshelves/College_Physics/Book%3A_College_Physics_\(OpenStax\)/02%3A_Kinematics/2.01%3A_Displacement](https://phys.libretexts.org/Bookshelves/College_Physics/Book%3A_College_Physics_(OpenStax)/02%3A_Kinematics/2.01%3A_Displacement)

Instantaneous velocity

- Instantaneous velocity is defined as the rate of change of position with respect to time.

$$\vec{v}_x = \lim_{\Delta t \rightarrow 0} \frac{\Delta \vec{x}}{\Delta t} = \frac{d\vec{x}}{dt}$$


Gradient of a position-time graph

- SI unit: meter/second (m/s).

[https://phys.libretexts.org/Bookshelves/College_Physics/Book%3A_College_Physics_\(OpenStax\)/02%3A_Kinematics/2.03%3A_Time_Velocity_and_Speed](https://phys.libretexts.org/Bookshelves/College_Physics/Book%3A_College_Physics_(OpenStax)/02%3A_Kinematics/2.03%3A_Time_Velocity_and_Speed)

Acceleration

- Acceleration is defined as the rate of change of velocity with respect to time.

$$\vec{a}_x = \lim_{\Delta t \rightarrow 0} \frac{\Delta \vec{v}_x}{\Delta t} = \frac{d\vec{v}_x}{dt}$$

Gradient of a velocity-time graph



- SI unit: meter/second² (m/s²).

[https://phys.libretexts.org/Bookshelves/College_Physics/Book%3A_College_Physics_\(OpenStax\)/02%3A_Kinematics/2.04%3A_Acceleration](https://phys.libretexts.org/Bookshelves/College_Physics/Book%3A_College_Physics_(OpenStax)/02%3A_Kinematics/2.04%3A_Acceleration)

Kinematics (in x-direction)

1D motion: an object is moving along a straight line.

Position: $\vec{r}(t) = x \hat{i}$

Velocity: $\vec{v}(t) = v_x \hat{i}$

$$v_x = \frac{dx}{dt} \quad \begin{array}{l} v_x > 0, \text{ direction } + \hat{i} \\ v_x < 0, \text{ direction } - \hat{i} \end{array}$$

Acceleration: $\vec{a}(t) = a_x \hat{i}$

$$a_x = \frac{dv_x}{dt} \quad \begin{array}{l} a_x > 0, \text{ direction } + \hat{i} \\ a_x < 0, \text{ direction } - \hat{i} \end{array}$$

[https://phys.libretexts.org/Bookshelves/University_Physics/Book%3A_Introductory_Physics_-_Building_Models_to_Describe_Our_World_\(Martin_Neary_Rinaldo_and_Woodman\)/03%3A_Describing_Motion_in_One_Dimension](https://phys.libretexts.org/Bookshelves/University_Physics/Book%3A_Introductory_Physics_-_Building_Models_to_Describe_Our_World_(Martin_Neary_Rinaldo_and_Woodman)/03%3A_Describing_Motion_in_One_Dimension)

Kinematics (in y -direction)

1D motion: an object is moving along a straight line.

Position: $\vec{r}(t) = y \hat{j}$

Velocity: $\vec{v}(t) = v_y \hat{j}$

$$v_y = \frac{dy}{dt} \quad \begin{array}{l} v_y > 0, \text{ direction } + \hat{j} \\ v_y < 0, \text{ direction } - \hat{j} \end{array}$$

Acceleration: $\vec{a}(t) = a_y \hat{j}$

$$a_y = \frac{dv_y}{dt} \quad \begin{array}{l} a_y > 0, \text{ direction } + \hat{j} \\ a_y < 0, \text{ direction } - \hat{j} \end{array}$$

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Case problem 1

- The position of a car is given as $x(t) = -t^2 + 3$ m, determine
 - a) its average velocity from $t = 0$ s to $t = 1$ s,
 - b) its instantaneous velocity at time $t = 1$ s.
- Are the two answers the same?

Case problem 1: Solution

- The position of a car is given as $x(t) = -t^2 + 3$ m, determine
 - a) its average velocity from $t = 0$ s to $t = 1$ s,
 - b) its instantaneous velocity at time $t = 1$ s.
- Are the two answers the same?

$$v_{x,avg} = \frac{\Delta x}{\Delta t} = \frac{2 - 3}{1 - 0} = -1 \text{ m/s}$$

$$v_x = \left. \frac{dx}{dt} \right|_{t=1} = -2t \Big|_{t=1} = -2 \text{ m/s}$$

Case problem 2

- The position of a car is given as $x(t) = t^3 + 3t^2$ m, determine
 - a) its instantaneous velocity at time t .
 - b) its instantaneous acceleration at time t .

Case problem 2: Solution

- The position of a car is given as $x(t) = t^3 + 3t^2$ m, determine
 - a) its instantaneous velocity at time t .
 - b) its instantaneous acceleration at time t .

$$v_x(t) = \frac{d}{dt} x(t) = 3t^2 + 6t$$

$$a_x(t) = \frac{d}{dt} v_x(t) = 6t + 6$$

Constant acceleration

$$v_x(t) = v_{x,0} + a_x t$$

$$x(t) = x_0 + v_{x,0}t + \frac{1}{2}a_x t^2$$

- For constant acceleration, acceleration along the x -axis = a_x

- Velocity, $v_x(t) = v_{x,0} + \int_0^t a_x dt = v_{x,0} + a_x t$

For your information only!

- Position, $x(t) = x_0 + \int_0^t v_x(t) dt = x_0 + v_{x,0}t + \frac{1}{2}a_x t^2$

- Note: these equations are only valid for constant acceleration

<https://openstax.org/books/university-physics-volume-1/pages/3-6-finding-velocity-and-displacement-from-acceleration>

Kinematics (in x-direction): constant acceleration

Position: $\vec{r}(t) = x \hat{i}$

Velocity: $\vec{v}(t) = v_x \hat{i}$

$$v_x = \frac{dx}{dt} \quad \begin{array}{l} v_x > 0, \text{ direction } + \hat{i} \\ v_x < 0, \text{ direction } - \hat{i} \end{array}$$

Acceleration: $\vec{a}(t) = a_x \hat{i}$

$$a_x = \frac{dv_x}{dt} \quad \begin{array}{l} a_x > 0, \text{ direction } + \hat{i} \\ a_x < 0, \text{ direction } - \hat{i} \end{array}$$

Acceleration: $\vec{a}(t) = a_x \hat{i}$

Velocity: $\vec{v}(t) = v_x \hat{i}$

$$v_x = v_{x,0} + a_x t \quad \begin{array}{l} v_x > 0, \text{ direction } + \hat{i} \\ v_x < 0, \text{ direction } - \hat{i} \end{array}$$

Position: $\vec{r}(t) = x \hat{i}$

$$x = x_0 + v_{x,0}t + \frac{1}{2}a_x t^2 \quad \begin{array}{l} x > 0, \text{ direction } + \hat{i} \\ x < 0, \text{ direction } - \hat{i} \end{array}$$

Case Problem 3

A drag racer starts from rest and accelerates at 7.4 m/s^2 . How far has it traveled in 2s? At what velocity?

Case Problem 3: Solution

A drag racer starts from rest and accelerates at 7.4 m/s^2 in x direction. How far has it traveled in 2s? At what velocity?

$$x = x_0 + v_{x,0}t + \frac{1}{2}a_x t^2 \rightarrow x(2) = \frac{1}{2} \times 7.4 \times 2^2 = 14.8 \text{ m}$$

$$v_x = v_{x,0} + a_x t \rightarrow v_x(2) = 7.4 \times 2 = 14.8 \text{ m/s}$$

Kinematics (in y-direction): constant acceleration

Position: $\vec{r}(t) = y \hat{j}$

Velocity: $\vec{v}(t) = v_y \hat{j}$

$$v_y = \frac{dy}{dt} \quad \begin{array}{l} v_y > 0, \text{ direction } + \hat{j} \\ v_y < 0, \text{ direction } - \hat{j} \end{array}$$

Acceleration: $\vec{a}(t) = a_y \hat{j}$

$$a_y = \frac{dv_y}{dt} \quad \begin{array}{l} a_y > 0, \text{ direction } + \hat{j} \\ a_y < 0, \text{ direction } - \hat{j} \end{array}$$

Acceleration: $\vec{a}(t) = a_y \hat{j}$

Velocity: $\vec{v}(t) = v_y \hat{j}$

$$v_y = v_{y,0} + a_y t \quad \begin{array}{l} v_y > 0, \text{ direction } + \hat{j} \\ v_y < 0, \text{ direction } - \hat{j} \end{array}$$

Position: $\vec{r}(t) = y \hat{j}$

$$y = y_0 + v_{y,0}t + \frac{1}{2}a_y t^2 \quad \begin{array}{l} y > 0, \text{ direction } + \hat{j} \\ y < 0, \text{ direction } - \hat{j} \end{array}$$

Gravitational Free Fall

- Choose coordinate system with y -axis vertical, and take upwards positive,
- Acceleration: $a_y = -g = -9.81 \text{ m} \cdot \text{s}^{-2}$
- Velocity: $v_y = v_{y,0} + a_y t = v_{y,0} - gt$
- Position: $y = y_0 + v_{y,0}t + \frac{1}{2}a_y t^2 = y_0 + v_{y,0}t - \frac{1}{2}gt^2$

https://en.wikipedia.org/wiki/Free_fall

Example

A ball is thrown straight upward with an initial velocity of 8.9 m/s. If the acceleration of the ball is -9.8 m/s^2 , find its velocity and position after 1 s.

Example: Solution

A ball is thrown straight upward with an initial velocity of 8.9 m/s. If the acceleration of the ball is -9.8 m/s^2 , find its velocity and position after 1 s.

$$v_y = v_{y,0} - gt \rightarrow v_y(1) = 8.9 + (-9.8) \times 1 = -0.9 \text{ m/s}$$

$$y = y_0 + v_{y,0}t - \frac{1}{2}gt^2 \rightarrow y(1) = 8.9 \times 1 - \frac{1}{2} \times 9.8 \times 1^2 = 4 \text{ m}$$

Example

A ball is vertically thrown upwards with an initial velocity v_0 . Determine the signs of its displacement, velocity and acceleration at different time points by taking upwards as positive.

Point	Position	Velocity	Acceleration
1	0	+ve	-ve
2			
3			
4			
5			
6			

3 ●

4 ● ↓

2 ● ↑

1 ● v_0 ● ↓ 5

6 ● ↓

[https://phys.libretexts.org/Bookshelves/College_Physics/Book%3A_College_Physics_\(OpenStax\)/02%3A_Kinematics/2.07%3A_Falling_Objects](https://phys.libretexts.org/Bookshelves/College_Physics/Book%3A_College_Physics_(OpenStax)/02%3A_Kinematics/2.07%3A_Falling_Objects)

Example: Solution

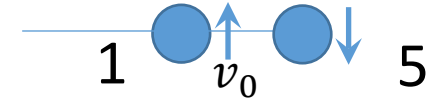
A ball is vertically thrown upwards with an initial velocity v_0 . Determine the signs of its displacement, velocity and acceleration at different time points by taking upwards as positive.

Point	Position	Velocity	Acceleration
1	0	+ve	-ve
2	+ve	+ve	-ve
3	+ve	0	-ve
4	+ve	-ve	-ve
5	0	-ve	-ve
6	-ve	-ve	-ve

3 ●

4 ● ↓

2 ● ↑


 1 ● v_0 ● 5

6 ● ↓

Case Problem 4:

You throw a ball vertically upward from the roof of a tall building. The ball leaves your hand at a point even with the roof railing with an upward velocity of 15.0 m/s ; the ball is then in free fall. On its way back down, it just misses the railing. Find (a) the ball's position and velocity 1.00 s and 4.00 s after leaving your hand; (b) the ball's velocity when it is 5.00 m above the railing; (c) the maximum height reached; (d) the ball's acceleration when it is at its maximum height. Take $g = 9.8 \text{ m/s}^2$.

Case Problem 4: Solution

You throw a ball vertically upward from the roof of a tall building. The ball leaves your hand at a point even with the roof railing with an upward velocity of 15.0 m/s; the ball is then in free fall. On its way back down, it just misses the railing. Find (a) the ball's position 1.00 s and 4.00 s after leaving your hand; (b) the ball's velocity when it is 5.00 m above the railing; (c) the maximum height reached; (d) the ball's acceleration when it is at its maximum height. Take $g = 9.8 \text{ m/s}^2$.

$$y = y_0 + v_{y,0}t - \frac{1}{2}gt^2 = 15t - \frac{1}{2}gt^2 \rightarrow y(1) = 15 \times 1 - \frac{1}{2} \times 9.8 \times 1^2 = 10.1 \text{ m}$$

$$y(4) = 15 \times 4 - \frac{1}{2} \times 9.8 \times 4^2 = -18.4 \text{ m}$$

Case Problem 4: Solution

You throw a ball vertically upward from the roof of a tall building. The ball leaves your hand at a point even with the roof railing with an upward velocity of 15.0 m/s; the ball is then in free fall. On its way back down, it just misses the railing. Find (a) the ball's position 1.00 s and 4.00 s after leaving your hand; (b) the ball's velocity when it is 5.00 m above the railing; (c) the maximum height reached; (d) the ball's acceleration when it is at its maximum height. Take $g = 9.8 \text{ m/s}^2$.

$$y = y_0 + v_{y,0}t - \frac{1}{2}gt^2 = 15t - \frac{1}{2}gt^2 \rightarrow y(t) = 15t - \frac{1}{2}gt^2 = 5 \text{ m} \rightarrow t = 0.38 \text{ or } 2.68$$

$$v_y = v_{y,0} - gt = 15 - gt \rightarrow v_y(0.38) = 15 - 9.8 \times 0.38 = 11.3 \text{ m/s}$$

$$v_y(2.68) = 15 - 9.8 \times 2.68 = -11.3 \text{ m/s}$$

H. D. Young and R. A. Freedman, University Physics (14th edition): Example 2.7, p75

Case Problem 4: Solution

You throw a ball vertically upward from the roof of a tall building. The ball leaves your hand at a point even with the roof railing with an upward velocity of 15.0 m/s; the ball is then in free fall. On its way back down, it just misses the railing. Find (a) the ball's position 1.00 s and 4.00 s after leaving your hand; (b) the ball's velocity when it is 5.00 m above the railing; (c) the maximum height reached; (d) the ball's acceleration when it is at its maximum height. Take $g = 9.8 \text{ m/s}^2$.

$$v_y = v_{y,0} - gt = 15 - gt \rightarrow v_y(T) = 0 = 15 - 9.8T \rightarrow T = 1.53$$

$$y = y_0 + v_{y,0}t - \frac{1}{2}gt^2 = 15t - \frac{1}{2}gt^2 \rightarrow y(T) = 15T - \frac{1}{2}gT^2 = 11.5 \text{ m}$$

H. D. Young and R. A. Freedman, University Physics (14th edition): Example 2.7, p75

Case Problem 4: Solution

You throw a ball vertically upward from the roof of a tall building. The ball leaves your hand at a point even with the roof railing with an upward velocity of 15.0 m/s ; the ball is then in free fall. On its way back down, it just misses the railing. Find (a) the ball's position 1.00 s and 4.00 s after leaving your hand; (b) the ball's velocity when it is 5.00 m above the railing; (c) the maximum height reached; (d) the ball's acceleration when it is at its maximum height. Take $g = 9.8 \text{ m/s}^2$.

A free-fall misconception: It is a common misconception that at the highest point of the motion, where the velocity is zero, the acceleration is also zero.

At every point, including the highest point, and at any velocity, including zero, the acceleration in free fall is always $a_y = -g$.

H. D. Young and R. A. Freedman, University Physics (14th edition): Example 2.7, p75